



Project Proposal On

"REDEFINE Robust and Explainable DEep neural networks for INfErrence"

Submitted to

Division :International Cooperation Division

Programme or Scheme : Indo-French Joint Research call by DST-ANR

Submitted by

Project Investigator:

Dr. Arijit Mondal

INDIAN INSTITUTE OF TECHNOLOGY PATNA-Daultpur

A. PROJECT IDENTIFICATION :		
1.	Title of Project	REDEFINE Robust and Explainable DEep neural networks for INfErrence
2.	Duration of Project	3 Years
3.	Research Field for cooperation	Mathematics for trustworthy and reliable AI
4.	Sub-field of Proposed Research Field	Interpretability and explainability of AI systems are mandatory so that solutions provided by such systems can be explained (to humans) Algebraic and formal modeling for AI, low rank matrix and tensor decomposition Algebraic techniques provide tools for analyzing and designing AI models with structured representations
5.	Key words qualifying the scope of the proposal	deep learning, robustness, explainability, Low-rank-approximation, orthogonal models.
6	Type of Proposal	Proposal Against Call
7	Proposal Submit Date	20/04/2026

8 Summary of Proposal :

Summary

9. Details of Indian Principle Investigators (PI) and Collaborating Institutions :		
1.	Name	Dr. Arijit Mondal
	Gender	Male
	Date of Birth	28/12/1979
	Designation	Assistant Professor
	Department	CSE
	Institute/University	INDIAN INSTITUTE OF TECHNOLOGY PATNA
	State	Bihar
	District	Patna
	City/Place	Daultapur
	Address	Bihta Kanpa Road
	Date of Superannuation	31/12/2044
	Pin	801106
	Communication Email	arijit@iitp.ac.in
	Alternate Email	amondal@gmail.com
	Mobile No	9632127613
	Phone	
	Fax	
	Category	General

10. Details of Indian Co-Principle Investigators (Co-PI) and Collaborating Institutions :		
1.	Name	Prof. Jimson Mathew
	Gender	Male
	Date of Birth	25/04/1971
	Designation	Professor
	Department	CSE
	Institute/University	INDIAN INSTITUTE OF TECHNOLOGY (IIT) PATNA
	State	Bihar
	District	Patna
	City/Place	Bihta
	Address	Dept of CSE
	Date of Superannuation	30/04/2036
	Pin	801106

	Communication Email	jimson@iitp.ac.in
	Alternate Email	mathew.jimson@gmail.com
	Mobile No	8987379924
	Phone	
	Fax	
	Category	General

11. Details of Foreign Principle Investigators and Researchers (PI) and Collaborating Institutions :

1.	Name	Dr. Suraj Kumar
	Gender	Male
	Designation	Junior researcher
	Department	Roma Team
	Institute/University	INRIA
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	Communication Email	suraj.kumar@inria.fr
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12. Details of Foreign Co-Principle Investigators (Co-PI) and Collaborating Institutions :

1.	Name	Dr. Loris Marchal
	Gender	Male
	Designation	Senior researcher
	Department	Roma Team
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	Country	France
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	Alternate Email	
	Mobile No	
	Phone	
	Fax	

Part B. TECHNICAL INFORMATION :

1.	Objectives of the Project
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	The primary objective of this project is to design and develop robust, explainable, and reliable deep learning models for inference under adversarial and distributional uncertainties, leveraging matrix analysis, spectral theory, and adaptive learning frameworks. We aim to develop robust deep neural network architectures capable of defending against state-of-the-art adversarial attacks on image data. The approach will utilize an ensemble of small, spectrally-constrained and orthogonal neural networks, where complementary behaviors across models mitigate individual vulnerabilities. By enforcing orthogonality, bounded spectral norms, and low condition numbers, the proposed models will limit perturbation amplification and improve stability. The ensemble strategy will further enhance fault tolerance, generalization, and attack resilience. To enhance model transparency, we propose to develop non-negative matrix and tensor factorization (NMF/NTF)-based methods for interpreting learned representations. These methods will enable decomposition of deep network parameters and feature maps into parts-based, low-rank, and semantically meaningful components, facilitating explainable feature attribution. Additionally, factorization will support model compression, dimensionality reduction, and efficient inference, reducing the complexity of large-scale architectures while preserving performance. We propose a novel large language model (LLM)-driven interpretability framework, where algebraic and statistical properties of network parameters—such as singular values, spectral norms, condition numbers, and covariance structures—are used as auxiliary inputs. The LLM will act as a reasoning and explanation engine, translating these matrix-derived metrics into human-interpretable insights, enabling diagnostic analysis, consistency checking, and robustness-aware explanations. To further enhance robustness, we aim to integrate a reinforcement learning (RL)-based decision and control module that dynamically guides inference. The RL agent will leverage feedback from robustness metrics, uncertainty estimates, and sensitivity analysis to perform adaptive evaluation, model selection, and confidence calibration. This enables multi-stage inference, where predictions are refined iteratively to ensure stability, reliability, and robustness under adversarial and uncertain conditions.
2.	Scientific and technical novelty of the proposed work The proposed project introduces a set of novel, integrated methodologies that advance the state of the art in robust and explainable deep learning, with a strong foundation in matrix analysis, spectral theory, and adaptive learning frameworks. We propose the design of ensembles of small, orthogonal deep neural networks with explicitly controlled spectral norms and condition numbers. Unlike conventional monolithic architectures, this modular approach leverages complementary decision boundaries across models, enabling one model to compensate for the vulnerabilities of others. By enforcing orthogonality and Lipschitz constraints, the framework inherently limits perturbation amplification, leading to improved adversarial robustness, numerical stability, and generalization. This represents a shift from single-model defenses to cooperative, matrix-theoretically grounded architectures. The project introduces non-negative matrix and tensor factorization (NMF/NTF) as a principled tool for interpreting deep neural networks. By decomposing weight matrices and feature tensors into low-rank, parts-based representations, the approach yields semantically meaningful and human-interpretable components. In addition to explainability, this factorization enables model compression, redundancy reduction, and efficient inference, bridging the gap between interpretability and computational efficiency—an area that remains underexplored in current literature. A key novelty lies in integrating large language models (LLMs) with algebraic analysis of neural network parameters. The proposed framework feeds the LLM with matrix-derived descriptors—including singular values, spectral distributions, condition numbers, and covariance structures—along with saliency maps and feature attributions. This enables the LLM to generate context-aware, robustness-informed explanations, moving beyond surface-level interpretations toward structured reasoning over model internals. Furthermore, the project explores fine-tuning LLMs using algebraic and saliency-guided supervision, a novel direction for enhancing explanation fidelity and consistency. The project proposes a reinforcement learning (RL)-based agent that dynamically orchestrates the inference process. Instead of relying on a single forward pass, the RL agent utilizes robustness metrics, uncertainty estimates, and sensitivity analyses to guide multi-stage evaluation, model selection, and decision refinement. This enables adaptive inference, where the system can re-evaluate, defer, or calibrate predictions before issuing a final verdict. Such a closed-loop, feedback-driven mechanism significantly enhances reliability, robustness, and trustworthiness, particularly in adversarial and uncertain environments.
3.	Justification for collaboration and brief information about national and international scenario in the proposed area of research

	The proposed collaboration between Indian Institute of Technology Patna and INRIA is strategically designed to leverage complementary expertise and address the interdisciplinary challenges of robust and explainable deep learning. The Indian team at IIT Patna brings substantial expertise in robustness of deep learning models, with prior work focusing on adversarial resilience, secure inference, and practical deployment in real-world scenarios. Their experience in developing and evaluating defense mechanisms against adversarial and distributional attacks provides a strong application-driven foundation for the project. In contrast, the INRIA team contributes deep theoretical strength in mathematical analysis of matrices and tensors, including spectral theory, matrix perturbation analysis, and tensor decomposition techniques. Their expertise enables rigorous formulation of stability guarantees, interpretability frameworks, and principled model design grounded in advanced mathematics. The convergence of these two domains—applied robustness and theoretical matrix analysis—is essential for addressing the emerging need for trustworthy AI systems. Robustness and interpretability are inherently multi-disciplinary problems, requiring insights from machine learning, linear algebra, optimization, and statistical analysis. The proposed collaboration uniquely positions the team to bridge this gap and develop a holistic framework that integrates theoretical guarantees with practical robustness and explainability. Furthermore, both teams have demonstrated strong research capabilities, evidenced by publications in leading international venues and successful execution of high-impact, funded research projects. Their prior experience in handling large-scale, collaborative research initiatives ensures effective coordination, knowledge exchange, and timely delivery of project outcomes. This collaboration is expected to foster bilateral knowledge transfer, combining algorithmic innovation with mathematical rigor, and enabling the development of novel methodologies that neither team could achieve independently. The partnership will also strengthen long-term research ties between India and France in the domain of trustworthy and reliable artificial intelligence, aligning with global priorities in AI safety and deployment.
3.1	National
	In India, research on robust and trustworthy artificial intelligence is steadily gaining momentum, with several academic and research institutions contributing to different aspects of the problem. Leading institutes such as Indian Institute of Science, the IITs, and IIITs are actively working on areas including adversarial machine learning, secure AI systems, and explainable AI. Several efforts across institutions focus on defensive mechanisms, model robustness, and interpretability, although these efforts are often fragmented across domains. At the policy level, there is a growing recognition of the importance of safe and reliable AI systems. Initiatives by government bodies such as NITI Aayog and programs under the Ministry of Electronics and Information Technology (MeitY) emphasize responsible AI, security, and trustworthiness, particularly for applications in healthcare, smart mobility, agriculture, and national security. With increasing deployment of AI in critical sectors, there is a strong push to address challenges related to adversarial robustness, data security, and explainability.
3.2	International
	Research on adversarial machine learning has attracted significant attention in recent years, often described as a “cat-and-mouse” game between attack and defense strategies. A wide range of adversarial attacks—such as Fast Gradient Sign Method (FGSM), Projected Gradient Descent (PGD), BIM, AutoAttack, and Carlini & Wagner Attack—have been proposed to expose vulnerabilities in deep learning models. In response, researchers have developed various defense mechanisms, including adversarial training, ensemble-based approaches, input transformation, and noise filtering techniques. Parallel efforts in model interpretability have led to methods such as Grad-CAM, saliency maps, and Integrated Gradients, which aim to provide insights into model decisions. Despite these advances, achieving truly robust and trustworthy deep learning systems remains an open challenge. Many defenses are effective only against specific attacks and often fail under adaptive or unseen adversarial strategies. Similarly, existing interpretability methods, while useful, may lack stability and consistency under perturbations. This gap is particularly concerning in safety-critical applications, where unreliable predictions can lead to severe consequences. Therefore, there is a pressing need to develop holistic frameworks that jointly address robustness and interpretability, ensuring dependable performance of deep learning models in real-world and high-risk environments.
3.3	Importance of the proposed project in the context of current status

	Adversarial vulnerabilities in deep learning models have become a critical concern. It is well known that adding small, human-imperceptible perturbations to input images can cause even highly accurate models to produce incorrect predictions. These attacks reveal fundamental weaknesses, particularly the high sensitivity and instability of current deep learning systems. The risks are especially severe in safety-critical applications such as autonomous driving, medical diagnosis, and patient monitoring in intensive care units, where incorrect predictions can lead to catastrophic consequences. As deep learning systems are increasingly deployed in real-world environments, ensuring their robustness, reliability, and trustworthiness is essential. Although training on large-scale datasets improves accuracy, it does not guarantee robustness against adversarial or distributional perturbations. Moreover, most state-of-the-art models operate as black boxes, providing little insight into their decision-making processes. This lack of interpretability reduces user trust and limits their adoption in high-stakes domains. The proposed project addresses these challenges by developing robust and explainable deep learning frameworks. It proposes ensembles of small, orthogonal neural networks that provide complementary strengths, improving resistance to adversarial perturbations. By incorporating spectral constraints and matrix-theoretic principles, the approach enhances stability and numerical conditioning. In addition, the project focuses on generating transparent and human-understandable explanations. Overall, the work aims to enable safe, reliable, and trustworthy AI systems for critical applications.
4.	Proposed Methodology Details

The project is structured around two tightly coupled verticals: (i) robustness of deep learning models under adversarial and distributional perturbations, and (ii) explainability through principled, matrix-based analysis. A unifying layer based on matrix metrics (spectral norms, singular values, condition numbers, and covariance structures) bridges these verticals, while a reinforcement learning (RL) engine provides adaptive control for reliable inference.

Robustness of deep learning models:

Ensemble of small orthogonal models - Conventional monolithic models often rely on a limited set of highly discriminative features, making them vulnerable to small, adversarial perturbations. To address this, we propose an ensemble of compact, orthogonally-constrained deep neural networks, designed to learn diverse and complementary feature subspaces. Each model will be trained with orthogonality constraints and bounded spectral norms, ensuring Lipschitz continuity and limiting perturbation amplification. This will improve numerical stability and gradient flow. We will explicitly promote representation diversity across models using decorrelation losses / subspace separation, so that failures of one model are compensated by others. Predictions will be aggregated using robust fusion strategies (e.g., confidence-weighted averaging, disagreement-aware voting, or learned gating), improving fault tolerance. Models will be evaluated against gradient-based (FGSM/PGD), optimization-based (C&W), and black-box attacks, along with distribution shift scenarios. Robustness is quantified using attack success rate, worst-case accuracy, and certified bounds (via spectral/Lipschitz estimates). This module delivers a modular, spectrally grounded architecture that is inherently more resistant to adversarial noise than single large models.

Matrix metric aware LLM based explanation generation - To ensure trustworthy inference, we will introduce a novel LLM-driven explainability layer that reasons over matrix-theoretic descriptors of the model. A set of matrix features will be extracted and the same will be used to fine tune the LLM. From each model (and the ensemble), we will compute singular values / spectral norms (stability, sensitivity), condition numbers (numerical conditioning), feature covariance / eigen-spectra (representation structure), saliency maps / attribution scores (input relevance). These descriptors, along with predictions and uncertainties, are fed into an LLM via structured prompts. We further explore fine-tuning with algebraic supervision, aligning explanations with spectral and perturbation-based evidence. The LLM will generate human-interpretable justifications, robustness diagnostics (e.g., "high sensitivity due to elevated spectral norm in layer k"), consistency checks across ensemble members. We will evaluate perturbation-stability of explanations, ensuring that explanations remain consistent under small input changes—crucial for reliable interpretability. This module will move beyond surface-level saliency to reasoned, matrix-aware explanations, tightly coupled with robustness.

RL engine for improving robustness - We will design a high-level RL agent that orchestrates inference to enhance robustness, reliability, and user-aligned performance. The agent will observe prediction confidence, disagreement across ensemble, spectral metrics, gradient-based sensitivity, and uncertainty estimates. The agent will select ensemble members, trigger re-evaluation (e.g., with input smoothing, augmentation, or alternative pathways), adjust thresholds (accept/reject/defer), invoke explanation checks via the LLM. We will develop a reward function that balances accuracy, adversarial robustness (worst-case loss), stability (low sensitivity), and explanation consistency. Moreover, the agent enables iterative decision-making, performing multiple evaluation rounds before issuing a final verdict when risk is high. Over time, the RL policy learns to allocate compute and model choices efficiently, improving robustness under varying attack strengths and data shifts.

Explainability:

Generation of explanations -- In this part, we focus on the explainable aspects of the deep neural networks.

Low-rank adaptation (LoRA) based analysis: LoRA is an efficient technique for fine-tuning large language models without updating all their parameters. It freezes the original weights and trains only a small number of additional parameters in their low-rank formats. We plan to apply this technique to classify images in two ways.

- * First, we train the network in a conventional way. After that, we divide it into two parts. In the first part, we keep the weights fixed; in the second part, we retrain the parameters in a low-rank format. The main idea behind our approach is that the first part extracts important features from the images, while the second, low-rank part makes the decision.

- * Similar to the first approach, we view the network as consisting of two parts. The first part detects important features from the images. The second part is composed of several LoRA adapters, some of which are selected based on the input image.

Gradient-weighted Class Activation Mapping (Grad-CAM) based analysis: Grad-CAM is a popular technique to visualize which parts of an image are important in the network to make a decision. We plan to focus on such parts. This approach will enable us to consider the explainability of small

	segments. Non-negative Matrix Factorization (NMF) based analysis: NMF decomposes a matrix into parts that are additive and non-negative. The results are relatively easy to interpret. We also plan to analyze weight matrices in deep neural networks using NMF. Similarity between models: Comparing two models, in general, is difficult. We plan to analyze two models based on Grad-CAM analysis. If the decisions are made based on the same segments, there are high chances that both models are similar. We also plan to analyze portions of two models with the help of low-rank and/or non-negative factorizations.
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4.1 Proposed Methodology the objectives and the country against each activity should be clearly mentioned :

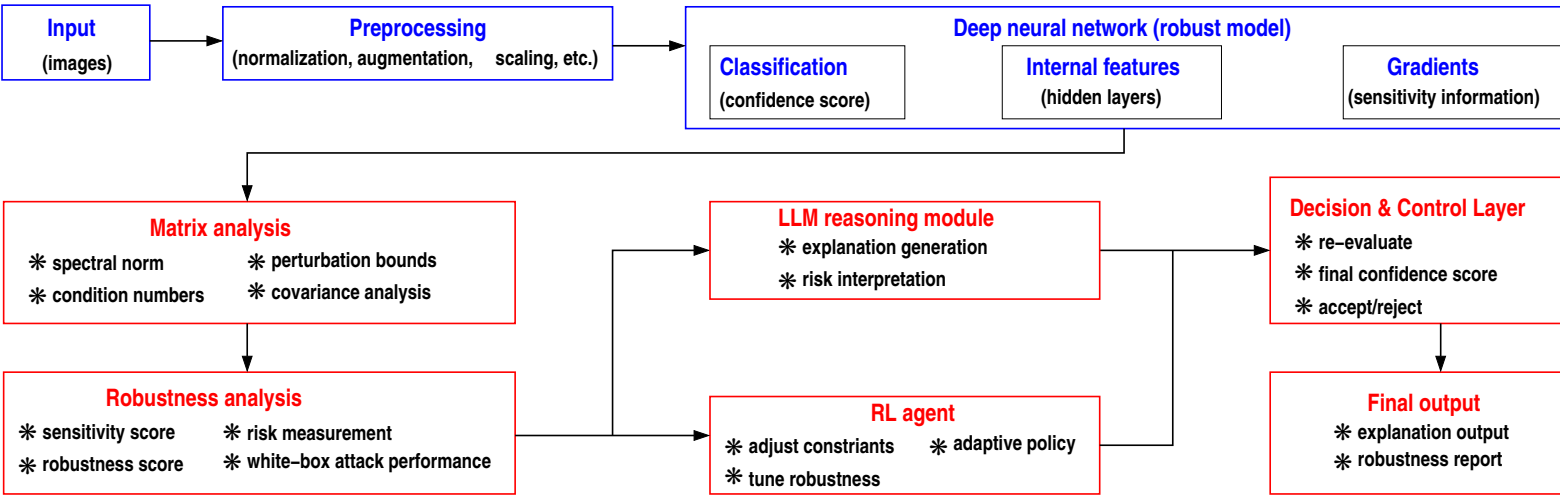
Time schedule	Objective	Relevant Activity	Responsibility (Indian or French Research team)
1 Year	1. Literature survey on adversarial attacks (FGSM, PGD, CW, etc.), robustness measures, LLM based explainability method, reinforcement learning based policy design 2. Design of small orthogonal models for building robust model, analysis of robustness measures 3. Development of LLM based pipeline for explainability generation taking account algebraic analysis outcomes, evaluation of explanation quality 4. Development of RL agent with reward function that considers accuracy, sensitivity, confidence score etc. for improving robustness 5. Perform a low-rank based analysis to understand which patterns are detected across different portions	Detailed plan for implementation, literature survey	Indian team (1,2,3,4), French team (5)

2 Year	1. Development of ensemble of orthogonal models, study & evaluation of complementarity and diversity metrics, evaluation against attacks such as PGD, AutoAttack, etc. 2. Fine tuning of LLMs using matrix metrics, saliency maps, heat maps, etc. Development of structured prompt for better explanation generation, comparison of explanation under perturbation 3. Integration of robustness metrics in RL engine, evaluation against adversarial attack, development of multi stage pipeline, adaptive re-evaluation strategy. 4. Use Grad-CAM based analysis to identify the parts that are important for making a decision. 5. Apply non-negative matrix/tensor factorization to the weight matrices to improve the explainability of the networks. Implementation of the proposed low-rank based networks on state-of-the-art hardware platforms.	Software development, code implementation, development of mathematical models	Indian team (1,2,3) French team (4,5)
3 Year	1. Rigorous testing of robust models against different attacks, optimization for efficient inference, final benchmarking 2. Development of robustness aware explanations, explainability engine, integration with other blocks 3. Combine RL with LLM feedback, full system integration, performance evaluation of the system 4. Full integration of all modules in a single platform 5. Study similarities between deep neural network models to understand their learned representations, architectural behaviors, and decision-making processes. 6. Scalability study of the low-rank based implementations on multiple GPUs.	Integration of different modules, final testing of software tools, documentation, report preparation	Indian team (1,2,3,4) French team (5,6)

4.2 Role of Indian and French teams details :

Team Country	Team Role Details
India	The Indian team will primarily focus on the robustness aspect of deep learning models. The team will plan to develop an ensemble of small orthogonal deep learning models. Small models can reduce computing resources and orthogonality of models can improve robustness of the model. The team will also build a LLM based methodology to provide explanations. The will also be responsible for development of an RL agent that can guide the user for better robustness through a series of evaluations.
French	The primary role for the French team will be development of mathematical models for the analysis of deep learning models. This will include low rank based analysis. It can help reduce the size of models as well as provide more insight about the model in terms of orthogonality, condition number, etc. Further, the team will also look into different schemes of interpretability models such as saliency map, grad-cam, etc. Both teams will collaborate with each other.

4.3 Uploaded figures/flowcharts/photographs (if any) referred in the project :



5. Plan of work along with Mapping of work :

S.N.	Time schedule	Country	Work Plan Details
1 .	1 Year	India	Literature survey, development of small models, evaluation for orthogonality measures, setting up LLM pipeline, development of RL engine
2 .	2 Year	India	Development of ensemble of orthogonal models, evaluation against attacks, algebraic analysis based fine tuning of LLM for explainability, multi-stage RL engine development and performance evaluation
3 .	3 Year	India	Full integration of different modules, rigorous testing of tool, performance evaluation, optimization for inference
4 .	1 Year	French	Literature survey on mathematical models for deep learning models, low rank based analysis
5 .	2 Year	French	Grad-CAM based analysis to identify the important parts, non-negative matrix/tensor factorization to the weight matrices, implementation of low-rank based analyzer
6 .	3 Year	French	Similarity analysis between learned representation of deep learning models and decision making process, scalability of low rank analysis

6.	Risk Management (max 200 words):(Identify, assess, and prioritise the most important risks in the project with reference to the listed milestones. Briefly describe the plan to mitigate the specific risks and reduce the project's overall risk profile.)
	The proposed project involves development of mathematical models and software. The risk associated with the project are as follows: Limited robustness improvement - In case orthogonal deep learning models may not yield expected robustness, we can characterize joint effects of multiple models. This can provide more insight to the problem. Such knowledge can provide us with more ideas how the models can be improved. Robustness vs accuracy tradeoff - Improving robustness may degrade clean accuracy. We will use multi-objective optimization and ensemble fusion strategies to balance accuracy–robustness trade-offs.
7.	Legal, ethical, or regulatory demands (max 150 words)
	The proposed project does not deal with any living beings. Therefore, we do not need any clearance for these.
8.	Value creation - growth and employment (max 100 words):
	Through this project we plan to develop mathematical models as well as software tools. Depending on the outcome we may explore commercialization of the software tools. Also, we may plan to come up with a more focused elective subject for the undergraduate, graduate students. We may also plan for a short term course on a similar domain.

9.	Expected results of this cooperation (e.g. joint publications, patents, etc. Please explain whether the expected results are likely to have commercial value and how intellectual property rights will be shared.)
	Through this project we expect that several joint publications will be made. There will be separate publications from the individual team also. Depending on the progress of the project we may plan for patents or commercializing the intellectual property. In such a case, the ownership will lie with both teams.
10.	Details of how intellectual property rights will be shared
	It will be owned by both institutes
11.	Facilities related to project activity available at the institutions where the project will be carried out:
a	At the Collaborating Indian Institutions:
	The Indian team will look into mostly the robustness aspect of the project. For which it will rely on orthogonal deep learning models. The Indian will develop an LLM based pipeline for explanation generation. They will also develop a multi-stage RL engine. Both teams will interact on a regular basis and contribute to all aspects.
b	At the Collaborating French Institutions:
	The French team will primarily be responsible for developing mathematical models. This will include a low rank approximation of the weight matrix/tensor. The team will analyze the models using matrix based methods and the extracted information will be propagated to LLM for more meaningful explanation generation.

C. ADMINISTRATIVE & FINANCIAL INFORMATION :

1.	1.Designation and address of the Authorized official in Indian Institute/ French Institute/ University to receive the grant
	IFSC Code : SBIN0017164
	Name of the Bank : State Bank of India
	Bank Branch Name : IIT BIHTA
	MICR Code : 801002005
	Account Type : Saving
	Name of A/C Holder : IIT Patna
	Bank A/C No : 30957551934

2. Project Cost :

(a) For the the French side (Euro) :

Budget Head	Year-1	Year-2	Year-3	Total
Any Other Recurring	0	3000	0	3000
Consumables Minor Equipment etc	2500	2500	0	5000
Exchange visits	16300	16600	16600	49500
Human Resource/ Manpower	42000	68000	68000	178000
Overhead	31800	0	0	31800
Total	92600	90100	84600	267300

Budget Head	Year-1	Year-2	Year-3	Total
1- Non-Recurring				
Equipment	3300000	0	0	3300000

Subtotal (Capital)	3300000	0	0	3300000
2- Recurring				
Project Staff	1512000	1512000	1512000	4536000
Contingency	50000	50000	50000	150000
Travel	435000	572500	720000	1727500
Overhead	1096400	443900	613400	2153700
Any Other Recurring	0	0	700000	700000
Subtotal (General)	3278400	2663400	3680400	9622200
Total Project Cost (Capital + General)	6578400	2663400	3680400	12922200

Equipment

S.	Equipments	Qty.	Justification	1 Year	Total
1 .	GPU server	1	To run computation intensive tasks	2500000	2500000
2 .	Jetson board	2	To run deep learning model on edge devices	200000	200000
3 .	Workstation	4	To develop code and software	600000	600000
Total				3300000	3300000

B. Recurring

Project Staff

S.	Project Staff	No.	Justification	1 Year	2 Year	3 Year	Total
1 .	Senior Research Fellow (SRF)	3	To develop code, software and test	1512000	1512000	1512000	4536000
Total				1512000	1512000	1512000	4536000

Manpower is needed to develop software tools, implementing code, and testing the same. At IIT Patna, we have requested for three manpowers. Their roles have been decided as well.

Consumables

S.	Items	Qty.	Justification	1 Year	2 Year	3 Year	Total
1 .	Arduino board	10	To run deep learning models in small devices	50000	0	0	50000
2 .	cables, minor spare parts,	1	reparing	15000	15000	15000	45000
3 .	Hard drive	1	Storage of data	25000	25000	25000	75000
4 .	Patch chord and other accessories	5	connections	20000	20000	20000	60000
5 .	printer cartridge	3	printing	15000	15000	15000	45000
6 .	Raspberry pi board	10	To run deep learning models on small devices	50000	0	0	50000
7 .	UPS battery	2	power supply	10000	10000	10000	30000
Total				185000	85000	85000	355000

Consumables include UPS battery, small edge devices and relevant accessories, printer cartridges, etc. Such will be needed from time to time. Small edge devices such as arduino / R Pi boards may be used to test small model performance.

Contingency

S.	Description	Justification	1 Year	2 Year	3 Year	Total
1 .	Contingency	Unforeseen expenditure	50000	50000	50000	150000
Total			50000	50000	50000	150000

Travel

S.	Description	Justification	1 Year	2 Year	3 Year	Total
1 .	Travel	Exchange visit, project presentation	0	0	720000	720000
2 .	Travel	Exchange visits, project presentation	0	572500	0	572500
3 .	Travel	Project presentation, travel to France, exchange visits	435000	0	0	435000
Total			435000	572500	720000	1727500

Overhead

S.	Description	Justification	1 Year	2 Year	3 Year	Total
1 .	Overhead	20%	1096400	443900	613400	2153700
Total			1096400	443900	613400	2153700

Any Other Recurring

S.	Description	Justification	1 Year	2 Year	3 Year	Total
1 .	Organizing workshop	Dissemination of knowledge towards the end of project	0	0	700000	700000
Total			0	0	700000	700000

3. Number of exchange visits required to achieve the project objectives :

(a) French to India :

Period	Number of visitors	Duration (Number of Days)	Details of visitors
1 Year	1	30	phd1
2 Year	1	30	phd2
3 Year	2	45	phd3 30 days & faculty 15 days

(b) India to French : :

Period	Number of visitors	Duration (Number of Days)	Details of visitors
1 Year	2	45	phd1 30 days & fac1 15 days
2 Year	2	60	phd1 45 days & fac2 15 days
3 Year	2	30	phd1 15 days & fac1 15 days

4. Equipment available with the Institute/ Group/ Department/ Other Institutes for the project (List of facilities being extended by parent institution(s) for the project implementation) :

S.N.	Equipment available with	Generic Name of the Equipment	Model, Make & year of purchase	Equipment Remarks
1 .	Pls Department	GPU server	HP, 2019	99% usage

Part D OTHER SOURCES OF SUPPORT

1 .	Is this research currently being supported by other sources?	No
2 .	Has this project been submitted to other agencies for financial support?	No

3. Please indicate name of at least 3-4 Indian senior scientists working in this field :

NO	NAME	INSTITUTION	TEL & POSTAL ADDRESS	E-MAIL
1	Sanjay Kumar Singh	INDIAN INSTITUTE OF TECHNOLOGY (B.H.U)	, , -	sks.cse@iitbhu.ac.in
2	Chetan Arora	INDIAN INSTITUTE OF TECHNOLOGY DELHI	, , -	chetan@cse.iitd.ac.in

3	C. V. Jawahar	INTERNATIONAL INSTITUTE OF TECHNOLOGY HYDERABAD	, , -	jawahar@iiit.ac.in
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8.	PFMS Details		
	PFMS Unique Code Available:	No	

National Projects:

S.N o.	Project Title	Sponsoring Agency	Budget	Status
1	Deep Learned Detecation and Classification of Multiple Intrusions Using WDM Intensity and Phase-Sensitive OTDR in Underwater Environment- Phase II	Defence Research and Development Organisation	8500000.00	Ongoing
2	A Smart revenue optimization framework for on-demand scheduling & routing of heterogeneous EVs	Other	7500000.00	Ongoing

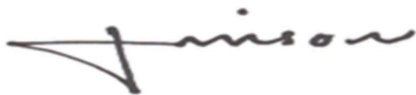
E. Certification :

Note: (i) The PI is required to submit periodic reports and its continuation in the second year would depend upon its progress assessed by DST or Ministry. At the end of each financial year. The PI is also required to submit the Utilization Certificates and Statement of Expenditure in prescribed format.

(ii) Visit of one person is to be counted as one visit.

(iii) For French applicants-The institute where the PI is employed should appoint at least one Co-PI .

I hereby Certify the statements above to be true and correct. I agree that DST has the right to reject the application if it is incomplete in any way.

Indian PI Signature	
French PI Signature	
Indian Co-PI Signature	
French Co-PI Signature	
Date	20/04/2026
Place	Bihta, Patna

List of Uploaded Documents:-

1. Declaration of PI / Co-PIs (As per given format, Annexure-I)
2. Declaration by the Heads of the Collaborating Institutions (As per given format, Annexure-II)
3. CV of the Indian PI (as per the given format, Annexure-III)
4. CV of the French PI (as per the given format, Annexure-III)
5. A signed certificate for the Conflict of Interest (as per the given format, Annexure-IV)
6. List of proposed Consumables and Accessories with price and make (attach as Annexure V).

CERTIFICATE FROM THE INVESTIGATOR

Project Title: **REDEFINE Robust and Explainable DEep neural networks for INfErence**

1. I/We agree to abide by the terms and conditions of the R&D grant.
2. I/We did not submit the same project proposal elsewhere for financial support.
3. I/We have explored and ensured that equipment and basic facilities (enumerated in the proposal) will actually be available as and when required for the purpose of the projects. I/We shall not request financial support under this project, for procurement of these items.
4. I/We undertake that spare time on permanent equipment (listed in the proposal) will be made available to other users.
5. I/We have enclosed the Endorsement from the Head of Institution.

Name and Signature of Investigator

Date...20.04.2026.....

Place.....Bihta & Lyon.....

SIGNATURES OF THE INDIAN AND FRENCH PRINCIPAL INVESTIGATORS



Arijit Mondal
INDIAN



Suraj Kumar
FRENCH



भारतीय प्रौद्योगिकी संस्थान पटना

INDIAN INSTITUTE OF TECHNOLOGY PATNA

डॉ अनूप कुमार केशरी
सह-अधिष्ठाता, अनुसंधान एवं विकास
Dr. Anup Kumar Keshri
Associate Dean, Research and Development

कनपारोड़, बिहटा, पटना- 801106, बिहार, भारत
Kanpa Road, Bihta, Patna – 801106, Bihar, India
Phone: +91-06115-233291
Email: adean_rnd@iitp.ac.in

Annexure-II

DECLARATION FROM THE HEADS OF THE COLLABORATING INSTITUTIONS

Project Title: REDEFINE: Robust and Explainable DEep neural networks for InfErrence

Indian members: Dr. Arijit Mondal, Prof. Jimson Mathew

French members: Dr. Suraj Kumar, Dr. Loris Marchal

It is certified that:

- (i) The Institutions agree to participate in this joint research project;
- (ii) The Institutions shall provide infrastructure and necessary facilities for im-plementing the joint project;
- (iii) The Institutions assume to undertake financial and other management re-sponsibilities for the part of the project work to be carried out at their insti-tution; and
- (iv) The backup funding for human resources, consumables, etc. is available to this project.
- (v) In the event of the Principal Investigator (PI) leaving the Institute due to re-tirement or otherwise, the Co-PI would become the PI and be responsible for completing the project and submitting the periodic progress report and financial statements;
- (vi) All necessary formalities as per our institution have been completed.

SIGNATURE & SEAL OF THE HEAD OF THE INSTITUTIONS

सह अधिष्ठाता
Associate Dean
अनुसंधान एवं विकास इकाई
Research & Development Unit
भारतीय प्रौद्योगिकी संस्थान पटना, बिहटा, पटना-801106
IIT Patna, Bihta, Patna-801106

INDIAN

FRENCH

BIO-DATA

1. Name: **Arijit Mondal**
2. Designation: Assistant Professor Grade I
3. Department: Dept of CSE
4. Institution: IIT Patna
5. Contact Number: 9632127613
6. Email: arijit@iitp.ac.in
7. Date of Birth: 28.12.1979
8. Gender: Male
9. Academic Qualification: (Undergraduate Onwards)

S.No.	Degree	Year	Subject	University/Institution	% of Marks
1	B.Tech	2001	EE	IIT Kharagpur	8.40
2	M.S.	2004	CSE	IIT Kharagpur	9.10
3	Ph.D.	2010	CSE	IIT Kharagpur	-

10. Ph.D. Thesis Title: A Symbolic Event Propagation Approach for Solving Timing Problems of Digital Circuits

Guide Name : Prof. P. P. Chakrabarti & Prof. Pallab Dasgupta

Institution : IIT Kharagpur

Year of Award : 2010

11. Work Experience (in chronological order)

S.No.	Position held	Institution	From	To
1	Senior Member of Technical Staff	Berkeley Design Automation	April, 2009	February, 2012
2	Member of Technical Staff	Cadence Design System	March, 2012	October, 2012
3.	Assistant Professor	IIT Patna	November, 2012	December, 2019
4.	Assistant Professor	CoEAI, IIT Kharagpur	December, 2019	July 2021
5.	Assistant Professor	IIT Patna	August 2021	Till now

12. Selected Publications (List of papers published in SCI Journals, in year wise descending order).

1. Adity Ghosh, Arnab Sarkar, Arijit Mondal, Dynamic Scheduling of Aperiodic Real-time Task Graphs on Heterogeneous Edge Computing Environments, ACM TECS 2025 (Accepted for publication)
2. Suraj Meshram, Arnab Sarkar, Arijit Mondal, A Tunable Generic Meta-Heuristic Framework for Balancing Assembly Line Systems in Manufacturing, To appear in ACM TECS, 2025
3. Dipankar Mandal, Arnab Sarkar, Arijit Mondal, A Discrete Partial Charging enabled Dynamic Programming Strategy for Optimal Fixed-Route Electric Vehicle Charging, To appear in ACM TECS, 2025
4. Ashiqur Rahaman Molla, Ram Mohan Chowdary Kota, Jaishree Mayank, Arnab Sarkar, Arijit Mondal, Soumyajit Dey, Minimizing Backbone Ethernet Traffic for Enabling Inter-zonal Messages in Software-Defined Vehicles, To appear in IEEE Embedded Systems Letters, 2025
5. Suraj Meshram, Sanket Jaipurkar, Arnab Sarkar, Arijit Mondal, An Efficient Iterative Beam Search for Human-Robot Collaborative Assembly Line Balancing, To appear in IEEE Embedded Systems Letters, 2025
6. Jyoti Kumari, Arijit Mondal, Jimson Mathew, Fourier-driven Lightweight Token Mixing Model for Efficient Time Series Forecasting, To appear in IEEE Transactions on Artificial Intelligence, 2025
7. Kavitha Yogaraj, Brian Quanz, Tarrun Vikas, Arijit Mondal, Samrat Mondal, Post-variational classical quantum transfer learning for binary classification, *Sci Rep* **15**, 23682 (2025). <https://doi.org/10.1038/>
8. Ranjit Kumar Mishra, Arijit Mondal, Jimson Mathew, Nystromformer based cross-modality transformer for visible-infrared person re-identification, Scientific Reports, Nature Publishing Group, May, 2025, Vol-15, Issue-1
9. Jaishree Mayank, Arijit Mondal, A Unified Job Scheduler for Optimization of Different System Performance Metrics, Concurrency and Computation: Practice and Experience, Wiley (accepted)
10. Anamika Dey, Somrita Sarkar, Arijit Mondal, and Pabitra Mitra. "Spatio-Temporal NDVI Prediction for Rice Crop." *SN Computer Science* 6, no. 3 (2025): 1-13.
11. Ranjit Kumar Mishra, Arijit Mondal, Jimson Mathew, Swin Transformer with Part-Level Tokenization for Occluded Person Re-identification, *Machine Vision and Applications* 36.1 (2025): 1-17.
12. Shivesh Kumar Sharan, Surajit Kumar Paul, Jyoti Kumari, and Arijit Mondal. "Artificial Neural Network-Based Prediction of Complete Forming Limit Curves for Steel in Sheet Metal Forming." *Journal of Alloys and Metallurgical Systems* (2025): 100166.
13. Shubhajit Dutta, Manaar Alam, Arijit Mondal, Debdeep Mukhopadhyay, Partha P Chakrabarti, Decision Guided Robust DL Classification of Adversarial Images Combining Weaker Defenses, IEEE Journal on Emerging and Selected Topics in Circuits and Systems. 2024
14. Somrita Sarkar, Anamika Dey, Ritam Pradhan, Upendra Mohan Sarkar, Chandranath Chatterjee, Arijit Mondal, and Pabitra Mitra, "Crop Yield Prediction Using Multimodal Meta-Transformer and Temporal Graph Neural Networks", Accepted for publication in IEEE Transactions on AgriFood Electronics, 2024
15. Suraj Singh, Ashiqur Rahaman Molla, Arijit Mondal, Soumyajit Dey, "Co-designing Perception-based Autonomous Systems on CPU-GPU platforms", *IEEE Embedded Systems Letters* (Accepted), 2024
16. Shruti Verma and Jyoti Kumari and Arijit Mondal and T. Santhanakrishnan and Jimson Mathew and Sumanta Gupta, Modeling and simulation of Φ -OTDR system suitable for underwater intrusion detection and classification, Optik, Elsevier, Vol-311, September 2024.
17. Surbhi Raj, Jimson Mathew, Arijit Mondal, Generalized and Robust Model for GAN-generated Image Detection, Pattern Recognition Letters, Elsevier, Volume 182, June 2024
18. Ritam Sarkar, Pranay Saha, Samrat Mondal, Arijit Mondal, MAGE: Microgrids with Advanced Grid Efficiency through Battery-Aware EV Management, Elsevier Sustainable Cities and Society, Vol 107, July, 2024
19. Sanghamitra Mishra, Arijit Mondal, Smart Mondal, Charging Station Siting and Sizing Considering Uncertainty in Electric Vehicle Charging Demand Distribution, IEEE Transactions of Intelligent Transportation Systems (2024 accepted for publication)

20. Pranay Saha, Nilotpal Chakraborty, Arijit Mondal, Samrat Mondal, Intelligent Real-time Utilization of Hybrid Energy Resources for Cost Optimization in Smart Microgrids, IEEE Systems journal, vol. 18, no. 1, pp. 186-197, March 2024
21. Jyoti Kumari, Jimson Mathew, Arijit Mondal, MAD-MEL: Combining Entity and Metric Learning for Anomaly Detection in Multivariate Time Series, IEEE Sensors journal, vol. 24, no. 3, pp. 3144-3156, 1 Feb.1, 2024
22. Surbhi Raj and Jimson Mathew and Arijit Mondal, FDT: A python toolkit for fake image and video detection, SoftwareX, Volume 22, May 2023, 101395
23. Fazail Amin, Arijit Mondal, Jimson Mathew, Person Re-Identification Using Selective Transformation Learning, Multimedia Tools and Applications, Springer, 2023.
24. Sangahmitra Mishra, Arijit Mondal, Samrat Mondal, A Multi-Objective Optimization Framework for Electric Vehicle Charge Scheduling with Adaptable Charging Ports, in IEEE Transactions on Vehicular Technology, doi: 10.1109/TVT.2022.3231901
25. Pranay Kumar Saha, Nilotpal Chakraborty, Arijit Mondal and Samrat Mondal, "Optimal Sizing and Efficient Routing of Electric Vehicles for a Vehicle-On-Demand System," IEEE Transactions on Industrial Informatics, vol. 18, no. 3, pp. 1489-1499, March 2022, doi: 10.1109/TII.2021.3091597
26. Fazail Amin, Arijit Mondal, and Jimson Mathew, "A Large Dataset With a New Framework for Abandoned Object Detection in Complex Scenarios," IEEE MultiMedia, May 2021
27. Nilotpal Chakraborty, Arijit Mondal and Samrat Mondal, "Intelligent Charge Scheduling and Eco-Routing Mechanism for Electric Vehicles: A Multi-Objective Heuristic Approach," Elsevier Sustainable Cities and Society, Vol 69, June 2021, 102820
28. Pranay Kumar Saha and Arijit Mondal, "Efficient Online Heuristic Approach for Handling Fluctuation in Renewable Energy in a Microgrid," IEEE Systems Journal, vol. 16, no. 2, pp. 2471 - 2482, June 2022
29. Nilotpal Chakraborty, Arijit Mondal and Samrat Mondal, "Efficient Load Control Based Demand Side Management Schemes Towards a Smart Energy Grid System," Elsevier Sustainable Cities and Society, vol. 59, 102175, August 2020.
30. Nilotpal Chakraborty, Arijit Mondal and Samrat Mondal, "Scheduling Interdependent Smart Appliances with Mixed-Preemption Policy for Peak Load Minimization," Taylor and Francis IETE Journal of Research, 66:6, 797-805, December 2020.
31. Basina D. R., Kumar S. , Padhi S. , Sarkar A. , Mondal A. , Kirthi R., "Brownout Based Blackout Avoidance Strategies in Smart Grids", IEEE Transactions on Sustainable Computing, vol. 6, no. 4, pp. 586-598, 1 Oct.-Dec. 2021, doi: 10.1109/TSUSC.2020.3014077.
32. Jaishree Mayank, and Arijit Mondal, "Efficient SAT Encoding Scheme for Schedulability Analysis of Non-preemptive Tasks on Multiple Computational Resources." Journal of Systems Architecture, June 2020.
33. Jana R. L., Dey S. , Dasgupta P. , Mondal A. "Automated Planning for finding Alternative Bug Traces", IET Computers & Digital Techniques, 14(6), 322-335, 2020.
34. Niraj Kumar, Jaishree Mayank and Arijit Mondal, "Reliability aware Energy Optimized Scheduling of Non-preemptive Periodic Real-Time Tasks on Heterogeneous Multiprocessor System," IEEE Transactions on Parallel and Distributed Systems, vol. 31, no. 4, pp. 871-885, 1 April 2020, doi: 10.1109/TPDS.2019.2950251.
35. Jaishree Mayank and Arijit Mondal, "Polynomial time schedulability test for periodic non-preemptive 2-task system," Information Processing Letters: Volume 154, February 2020.
36. Nilotpal Chakraborty, Arijit Mondal and Samrat Mondal, "Multi-Objective Optimal Scheduling Framework for HVAC Devices in Energy Efficient Buildings," IEEE Systems Journal, vol. 13, no. 4, pp. 4398-4409, Dec. 2019
37. Jaishree Mayank, Arijit Mondal and Arnab Sarkar, "Control-schedule co-design for fast stabilization in real time systems facing repeated reconfigurations," Design Automation for Embedded Systems, Springer, 2019
38. Jaishree Mayank and Arijit Mondal, "An Integer Linear Programming Framework for Energy Optimization of Non-Preemptive Real-Time Tasks on Multiprocessors," Journal of Low Power Electronics, vol. 15, no. 2, pp: 162–167, June 2019.

39. Niraj Kumar, Arijit Mondal, "Timing Analysis of Precedence Constraint Messages Scheduled over Dynamic Segment of FlexRay," IEEE Transactions on Automation Science and Engineering, vol. 17, no. 1, pp. 222-236, Jan. 2020, doi: 10.1109/TASE.2019.2907959
40. Ananya Singla, Varsha Agarwal, Sudip Roy and Arijit Mondal, "Reliability Analysis of Mixture Preparation using Digital Microfluidic Biochips," IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems, vol. 38, no. 4, pp. 654-667, April 2019.
41. Nilotpal Chakraborty, Arijit Mondal and Samrat Mondal, "Efficient Scheduling of Non-Preemptive Appliances for Peak Load Optimization in Smart Grid," in IEEE Transactions on Industrial Informatics, vol. 14, no. 8, pp. 3447-3458, Aug. 2018.
42. Nilotpal Chakraborty, Arijit Mondal and Samrat Mondal, "Intelligent Scheduling of Thermostatic Devices for Efficient Energy Management in Smart Grid," in IEEE Transactions on Industrial Informatics, vol. 13, no. 6, pp. 2899-2910, Dec. 2017.
43. Arijit Mondal, P. P. Chakrabarti, Pallab Dasgupta, "Symbolic Event Propagation Based Minimal Test Set Generation for Robust Path Delay Faults", ACM Transactions on Design Automation of Electronic Systems, Volume 17 Issue 4, October 2012.
44. Arijit Mondal, P. P. Chakrabarti, Pallab Dasgupta, "Statistical static timing analysis using symbolic event propagation", IET Circuit, Device and Systems, Vol 1, No 4, pages 283-291, 2007.
45. Arijit Mondal, P. P. Chakrabarti, "Reasoning about timing behavior of digital using symbolic event propagation and temporal logic", IEEE Transaction on Computer Aided Design of Integrated Circuits and Systems, Vol 25, No. 9, pages 1793-1814, 2006.

13. List of Projects implemented

Projects	Funding agency	Status	Budget, Role
Smart Energy Management in Office Building	IIT Patna	Completed	12L, PI
A Software Tool for the Planning and Design of Smart Micro Power Grids	IMPRINT-I	Completed	202L, Co-PI
Special Manpower Development Program - Chip to System Design (SMDP-C2SD)	MeitY	Completed	100L, Co-PI
Design and FPGA prototyping of multicarrier multiple access schemes for variable rate multimedia satellite communication	MeitY	Completed	90L, Co-PI
Development of Planning and Designing Tool for Smartly Adopting Electric Vehicles in Indian Cities	IMPRINT-II	Completed	63L, Co-PI
V2D: Video-to-Description Generation using Deep Learning	DST	Completed	32L, PI
Deep Learned Detection and Classification of Multiple Intrusions Using WDM Intensity and Phase-Sensitive OTDR in Underwater Environment	NRB, DRDO	Completed	46L, PI
A Smart revenue optimization framework for on-demand scheduling & routing of heterogeneous EVs	TIH, IIT Bhilai	Ongoing	75L, PI
AI for Physical Marketplace Management	RAKSHAK, DST	Completed	10L, PI
AI Based Digital Twin Support System for Mobility Analytics	IIT Kharagpur	Completed	25L, PI
Campus Rakshak: An Agent-based Simulation Framework for Assessment of Intervention Strategy against Covid-19 Spreading in Campuses	RAKSHAK, DST	Completed	10L, Co-PI

AI for Agriculture and Food Sustainability	DST	Completed	131L, Co-PI
AWS:NAIRP:Creating a Collateral for Supporting Popular ML Applications on AWS Infrastructure through National AI Resource Platform	Amazon	Completed	90L, Co-PI
IIT Patna Centre of Excellence in Cyber Crime Prevention against Women and Children -AI-based Tools for Women and Child Safety	BPRD, MHA	Completed	283L, Co-PI
Deep Learned Detecation and Classification of Multiple Intrusions Using WDM Intensity and Phase-Sensitive OTDR in Underwater Environment-Phase II	NRB, DRDO	Ongoing	85L, Co-PI

Suraj KUMAR (Inria Lyon, France)

1. Name and full correspondence address

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46 Allée d'Italie
69364 Lyon
France

2. Email(s) and contact number(s)

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3. Designation and Institution

Inria Starting Faculty Position (ISFP) researcher
Institut national de recherche en sciences et technologies du numérique (Inria) Lyon, France

4. Date of Birth 20-Nov-1989

5. Gender Male

6. Whether differently abled No

7. Academic Qualification PhD

8. Work experience (in chronological order)

Name and address of employer	Position held (Temporary/Regular)	Duration		Nature of Duties
		From	To	
Inria Lyon, France	Regular	Oct 2022	Ongoing	Research, teaching
Inria Paris, France	Temporary	Nov 2019	Sept 2022	Research
Pacific Northwest National Laboratory, USA	Temporary	May 2018	Oct 2019	Research

9. Professional Recognition/ Award/ Prize/ Certificate, Fellowship received by the applicant

S.No.	Name of Award	Awarding Agency	Year
1	Student travel awards	SC, IPDPS	2016

10. Publications (*I generally consider conferences to be my primary publication venues; however, for work that has greater rigor or deeper analysis, I prefer to publish in journals. The author names in publications generally appear in alphabetical order. List of papers published in SCI Journals, in year wise descending order during last 10 years*).

S.No	Author (s)	Title	Name of Journal	Volume	Page	Year
1	Hussam Al Daas, Grey Ballard, Laura Grigori, Suraj Kumar, Kathryn Rouse, and Mathieu Verite	Communication Lower Bounds and Optimal Algorithms for Symmetric Matrix Computations	ACM Transactions on Parallel Computing	Volume 12, Issue 2, Article No. 6	1-46	2025
2	Hussam Al Daas, Grey Ballard, Laura Grigori, Suraj Kumar, and Kathryn Rouse	Communication Lower Bounds and Optimal Algorithms for Multiple Tensor-Times-Matrix Computation	SIAM Journal on Matrix Analysis and Applications	Volume 45, Number 1	450-477	2024
3	Olivier Beaumont, Lionel Eyraud-Dubois, and Suraj Kumar	Fast approximation algorithms for task-based runtime systems	Concurrency and Computation: Practice and Experience	Volume 30, Number 17	e4502	2018

11. Conference Publications

S. No.	Title	Author's name	Conference name	Year of Publication
1	Brief Announcement: Minimizing Communication for Parallel Symmetric Tensor Times Same Vector Computation	Hussam Al Daas, Grey Ballard, Laura Grigori, Suraj Kumar, Kathryn Rouse, and Mathieu V�rit�	Symposium on Parallelism in Algorithms and Architectures (SPAA '25)	2025
2	Parallel Memory-Independent Communication Bounds for SYRK	Hussam Al Daas, Grey Ballard, Laura Grigori, Suraj Kumar, and Kathryn Rouse	Symposium on Parallelism in Algorithms and Architectures (SPAA '23)	2023
3	Brief Announcement: Tight Memory-Independent Parallel Matrix	Hussam Al Daas, Grey Ballard, Laura Grigori, Suraj	Symposium on Parallelism in Algorithms and	2022

	Multiplication Communication Lower Bounds	Kumar, and Kathryn Rouse	Architectures (SPAA '22)	
4	Analysis of a List Scheduling Algorithm for Task Graphs on Two Types of Resources	Lionel Eyraud-Dubois and Suraj Kumar	IEEE International Parallel and Distributed Processing Symposium (IPDPS '20)	2020
5	Performance Models for Data Transfers: A Case Study with Molecular Chemistry Kernels	Suraj Kumar, Lionel Eyraud-Dubois, and Sriram Krishnamoorthy	International Conference on Parallel Processing (ICPP '19)	2019
6	Approximation Proofs of a Fast and Efficient List Scheduling Algorithm for Task-Based Runtime Systems on Multicores and GPUs	Olivier Beaumont, Lionel Eyraud-Dubois, and Suraj Kumar	IEEE International Parallel and Distributed Processing Symposium (IPDPS '17)	2017
7	Scheduling of Linear Algebra Kernels on Multiple Heterogeneous Resources	Olivier Beaumont, Terry Cojean, Lionel Eyraud-Dubois, Abdou Guermouche, and Suraj Kumar	IEEE International Conference on High Performance Computing (HiPC '16)	2016

12. Languages Known

Language	Read (Excellent/Good/Fair)	Write (Excellent/Good/Fair)	Speak (Excellent/Good/Fair)
English	Excellent	Excellent	Excellent
French	Good	Good	Fair

13. Current research Activities (**Ongoing projects**)

- Parallelization of sketching based low-rank matrix factorizations
- Parallel algorithms for symmetric tensor decompositions
- Efficient GPU implementations of tensor algorithms

14. Partnerships and International Collaborations

I have close ties with EPFL, in particular with Laura Grigori, and with Wake Forest University, USA, in particular with Grey Ballard, both of whom are experts in designing efficient algorithms for low-rank approximations.

Declaration on Conflict of Interest

I have read the above “Policy on Conflict of Interest” of the DST applicable to the Applicant and agree to abide by provisions thereof.

I hereby declare that I have no conflict of interest of any form pertaining to the proposed grant* ✓

I hereby declare that I have conflict of interest of any form pertaining to the proposed grant* *

A handwritten signature in black ink that reads "Arijit Mondal". The script is cursive and fluid.

Arijit Mondal
INDIAN

List of consumables

Sr.No.	Year	Consumables Details	Justification	Qty	Amount	Total
1	Year 1	Raspberry pi board	To run deep learning models on small devices	10	5000.00	50000.00
2	Year 1	Arduino board	To run deep learning models in small devices	10	5000.00	50000.00
3	Year 1	UPS battery	power supply	2	5000.00	10000.00
4	Year 1	printer cartridge	printing	3	5000.00	15000.00
5	Year 1	Patch cord and other accessories	connections	5	4000.00	20000.00
6	Year 1	Hard drive	Storage of data	1	25000.00	25000.00
7	Year 1	cables, minor spare parts,	repairing	1	15000.00	15000.00
8	Year 2	UPS battery	power supply	2	5000.00	10000.00
9	Year 2	printer cartridge	printing	3	5000.00	15000.00
10	Year 2	Patch cord and other accessories	connections	5	4000.00	20000.00
11	Year 2	Hard drive	Storage of data	1	25000.00	25000.00
12	Year 2	cables, minor spare parts,	repairing	1	15000.00	15000.00
13	Year 3	printer cartridge	printing	3	5000.00	15000.00
14	Year 3	UPS battery	power supply	2	5000.00	10000.00
15	Year 3	Patch cord and other accessories	connections	5	4000.00	20000.00
16	Year 3	Hard drive	Storage of data	1	25000.00	25000.00
17	Year 3	cables, minor spare parts,	repairing	1	15000.00	15000.00